

Muhammad Musa

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ACADEMICS

University of Wisconsin - Madison – Masters Computer Sciences *September 2023 – December 2025*
Key Courses: ML, Language Security, Algorithms, Computer Vision, Privacy in Data, Big Data Systems, Data Mgmt, Foundation Models
Lahore University of Management Sciences (LUMS) – B.Sc. Computer Science CGPA: 3.9/4.0 *September 2019 – May 2023*
Key Courses: Database, Linear Algebra, Stats, Deep Learning, Data Mining, NLP, Network Security, Software Engineering, Signals

PROFESSIONAL EXPERIENCE

University of Wisconsin - Madison **Madison, WI**
Head Graduate Teaching Assistant for “CS 320: Data Science Programming II” *July 2024 – Present*

- Develop and grade assignments for intermediate topics in Python programming including **Git**, **OOP**, **Flask**, and Machine Learning.
- Lead a teaching staff of **20** to coordinate course content delivery and facilitate the learning of **600** students.

Graduate Teaching Assistant for “CS 220: Data Science Programming I” *August 2023 – May 2024*

SAUDCONSULT Saudi Consulting Services **Riyadh, Saudi Arabia**
Software Engineering Intern *June 2024 – September 2024*

- Developed architectural design tools using **AutoCAD API**, allowing automated structure data analysis and 3D modeling processes.
- Contributed to the integration of BIM (Building Information Modeling) technology with existing design platforms.
- Actively participated in daily stand-up meetings and bi-weekly sprints within Agile development framework with product team.

Lahore University of Management Sciences **Lahore, Pakistan**
Head Teaching Assistant for “CS 331: Introduction to Artificial Intelligence” *January 2022 – May 2023*

Research Intern and Teaching Assistant for “CS 3812: Introduction to Blockchain” *June 2021 – December 2021*

- Collaborated with **the university faculty** to architect course content on concepts like **smart contracts**, **Solidity**, and **NFTs**.
- Managed a diverse **inaugural** cohort of approximately 120 undergraduate and **graduate** students.

KASB Securities **Remote, Pakistan**
Smart Contract Developer *July 2021 – October 2021*

- Led a project to mint educational certificates as **NFTs** on the **STACKS blockchain**, using the language **Clarity** for smart contracts.
- Demonstrated innovation in blockchain development, contributing to enhancing data integrity and accessibility.

TransformX **Remote, Pakistan**
Web Developer *July 2021 – October 2021*

- Orchestrated the development and integration of web pages for a Communication Platform as a Service (**CPaaS**) organization.
- Crafted responsive, performance-optimized interfaces to meet the demanding requirements of CPaaS solutions.

RESEARCH EXPERIENCE

Data Creation and Active Learning for Low-Resource Languages – Paper in works *October 2024 – December 2024*

- Fine-tuned **XLM-RoBERTa** for topic classification, achieving **92.9% accuracy**, establishing a new benchmark for Shahmukhi NLP.
- Leveraged **active learning**, reducing labeled data requirements by **5 times**, optimizing resource efficiency.

LLM Jailbreaking & Security furthering Tree of Attacks Pruning (TAP) with Dr. Somesh Jha *January 2024 – April 2024*

- Developed a multi-evaluator jailbreak framework leveraging **GPT-3.5 Turbo**, refining jailbreaking scoring to reduce false positives and boost jailbreak success from **the current 4% to 68% with fewer queries** than **GPT-4**.

Ransomware Analysis with Dr. Ashish Gehani - Paper published in **ACM CSET '23** *August 2022 – August 2023*

- **Automated** the end-to-end pipeline, encompassing the execution of ransomware within virtual environments, real-time monitoring using **Process Monitor**, and generation of XML files for input into tools like **SPADE** for further analysis.

House Boundary Detection *March 2022 – May 2022*

- Employed the **VGG16 U-Net** architecture to perform high-precision boundary detection on low-resolution satellite images.

Sentiment Analysis – Poster published in **ASONAM '22** ([Link to poster](#)) *December 2021 – May 2022*

- Built a context-aware framework for extremist content detection and prevention through urban sentiment analysis on Twitter.

LEADERSHIP & COMMUNITY INVOLVEMENT

Project head for **TUBITAK TEKNOFEST UAV Competition**. Selected among the **top 20 teams globally**. *March 2021 – July 2021*

President, IEEE LUMS Student Chapter *August 2022 – June 2023*

Assistant Director HR, LUMS Community Service Society *August 2021 – June 2022*

SKILLS & SOFTWARES

Technical: Python (NumPy, Skicit-learn, PyTorch, TensorFlow, Pandas, StatsModels, Matplotlib, Keras), C++, C, R, MATLAB, Octave, Git, Docker, JavaScript, SQL, Solidity, Latex, WordPress, Google Cloud Platform, QGIS, Android Studio, SolidWorks, Scala, CI/CD, cloud computing, SAM, Detectron 2, GPT API, Hugging Face, Agile Framework, Large Language Models, Kubernetes, Github, Postman

Skills: critical thinking, problem-solving, communication skills